Markscheme

Mathematics: analysis and approaches
Practice question bank

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57. (a)

Consider the equation $e^{2x} - 7e^x + 12 = 0$.

Find the sum of the two values of x satisfying the equation, correct to 3 significant figures.

let $u = e^x$: $u^2 - 7u + 12 = 0$ *(M1)* **A1**

$(u - 3)(u - 4) = 0 \Rightarrow u = 3$ or $u = 4$ *(M1)*

$x = \ln 3$ or $x = \ln 4$ **A1**

sum $= \ln 3 + \ln 4 = \ln 12 \approx 2.48$ **A1**

[5 marks]